

Name : Saranya Aluru

Roll No: 2520080012

```
mysql> USE JoinPractice;
Database changed
mysql> CREATE TABLE class (
  ->     id INT,
  ->     name VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.08 sec)

mysql> CREATE TABLE class_info (
  ->     id INT,
  ->     address VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO class VALUES
  -> (1,'abhi'),
  -> (2,'adam'),
  -> (3,'alex'),
  -> (4,'anu');
Query OK, 4 rows affected (0.04 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO class_info VALUES
  -> (1,'DELHI'),
  -> (2,'MUMBAI'),
  -> (3,'CHENNAI');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT *  
      -> FROM class  
      -> CROSS JOIN class_info;
```

id	name	id	address
1	abhi	3	CHENNAI
1	abhi	2	MUMBAI
1	abhi	1	DELHI
2	adam	3	CHENNAI
2	adam	2	MUMBAI
2	adam	1	DELHI
3	alex	3	CHENNAI
3	alex	2	MUMBAI
3	alex	1	DELHI
4	anu	3	CHENNAI
4	anu	2	MUMBAI
4	anu	1	DELHI

12 rows in set (0.00 sec)

```
mysql> SELECT *  
      -> FROM class  
      -> INNER JOIN class_info  
      -> ON class.id=class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI

3 rows in set (0.00 sec)

```
mysql> SELECT class.name,  
->      class_info.address  
-> FROM class  
-> INNER JOIN class_info  
-> ON class.id=class_info.id;
```

```
+-----+-----+  
| name | address |  
+-----+-----+  
| abhi | DELHI   |  
| adam | MUMBAI  |  
| alex | CHENNAI |  
+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> SELECT *  
-> FROM class  
-> NATURAL JOIN class_info;
```

```
+-----+-----+-----+  
| id  | name | address |  
+-----+-----+-----+  
| 1   | abhi | DELHI   |  
| 2   | adam | MUMBAI  |  
| 3   | alex | CHENNAI |  
+-----+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> INSERT INTO class VALUES
-> (5,'ashish');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO class_info VALUES
-> (7,'NOIDA'),
-> (8,'PANIPAT');
Query OK, 2 rows affected (0.02 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id=class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL

5 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id=class_info.id
-> WHERE class_info.id IS NULL;
```

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL

```
mysql> SELECT *  
      -> FROM class  
      -> RIGHT JOIN class_info  
      -> ON class.id=class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

5 rows in set (0.00 sec)

```
mysql> SELECT *  
      -> FROM class  
      -> RIGHT JOIN class_info  
      -> ON class.id=class_info.id  
      -> WHERE class.id IS NULL;
```

id	name	id	address
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

2 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id=class_info.id
->
-> UNION
->
-> SELECT *
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id=class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

7 rows in set (0.01 sec)

```
mysql> SELECT *  
-> FROM class  
-> LEFT JOIN class_info  
-> ON class.id=class_info.id  
-> WHERE class_info.id IS NULL  
->  
-> UNION  
->  
-> SELECT *  
-> FROM class  
-> RIGHT JOIN class_info  
-> ON class.id=class_info.id  
-> WHERE class.id IS NULL;
```

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

4 rows in set (0.00 sec)

```
mysql> CREATE TABLE first_table(  
->     id INT,  
->     name VARCHAR(30)  
-> );  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> CREATE TABLE second_table(  
->     id INT,  
->     name VARCHAR(30)  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> INSERT INTO first_table VALUES  
-> (1,'abhi'),  
-> (2,'adam');  
Query OK, 2 rows affected (0.02 sec)  
Records: 2  Duplicates: 0  Warnings: 0  
  
mysql> INSERT INTO second_table VALUES  
-> (2,'adam'),  
-> (3,'chester');  
Query OK, 2 rows affected (0.02 sec)  
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT *  
-> FROM first_table  
->  
-> UNION  
->  
-> SELECT *  
-> FROM second_table;
```

id	name
1	abhi
2	adam
3	chester

```
3 rows in set (0.00 sec)
```



```
mysql> SELECT name
-> FROM first_table
->
-> UNION
->
-> SELECT name
-> FROM second_table;
```

```
+-----+
```

```
| name |
```

```
+-----+
```

```
| abhi |
```

```
| adam |
```

```
| chester |
```

```
+-----+
```

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM first_table
->
-> UNION ALL
->
-> SELECT *
-> FROM second_table;
```

```
+-----+-----+
```

```
| id  | name |
```

```
+-----+-----+
```

```
| 1  | abhi |
```

```
| 2  | adam |
```

```
| 2  | adam |
```

```
| 3  | chester |
```

```
+-----+-----+
```

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT COUNT(*)
-> FROM
-> (
-> SELECT *
-> FROM first_table
->
-> UNION ALL
->
-> SELECT *
-> FROM second_table
-> ) AS A;
```

```
+-----+
| COUNT(*) |
+-----+
|         4 |
+-----+
1 row in set (0.02 sec)
```

```
mysql> SELECT *
-> FROM first_table
-> INNER JOIN second_table
-> USING(id,name);
```

```
+-----+-----+
| id    | name  |
+-----+-----+
|      2 | adam  |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT first_table.name
-> FROM first_table
-> INNER JOIN second_table
-> ON first_table.name=second_table.name;
```

```
+-----+
| name |
+-----+
| adam |
+-----+
```

1 row in set (0.00 sec)

```
mysql> SELECT *
-> FROM first_table
-> WHERE id NOT IN
-> (
-> SELECT id
-> FROM second_table
-> );
```

```
+-----+-----+
| id  | name |
+-----+-----+
|  1  | abhi |
+-----+-----+
```

1 row in set (0.02 sec)

```
mysql> SELECT name
-> FROM first_table
-> WHERE name NOT IN
-> (
-> SELECT name
-> FROM second_table
-> );
```

```
+-----+
| name |
+-----+
| abhi |
+-----+
```

1 row in set (0.00 sec)

```
mysql> SELECT c.id,
->          c.name,
->          ci.address
-> FROM class c
-> INNER JOIN class_info ci
-> ON c.id=ci.id;
```

```
+-----+-----+-----+
| id  | name | address |
+-----+-----+-----+
|  1  | abhi | DELHI   |
|  2  | adam | MUMBAI  |
|  3  | alex | CHENNAI |
+-----+-----+-----+
```

3 rows in set (0.00 sec)

```
mysql> SELECT c.id,  
->         c.name,  
->         CASE  
->           WHEN ci.address IS NULL  
->           THEN 'Address Missing'  
->           ELSE 'Address Available'  
->         END AS Status  
-> FROM class c  
-> LEFT JOIN class_info ci  
-> ON c.id=ci.id;
```

id	name	Status
1	abhi	Address Available
2	adam	Address Available
3	alex	Address Available
4	anu	Address Missing
5	ashish	Address Missing

5 rows in set (0.00 sec)